

Package ‘dwf4r’

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Type Package

Title Unofficial R Interface to the Digilent WaveForms SDK

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Description An independently developed R interface to selected functions of the Digilent WaveForms SDK and its DWF library. The current version supports only a limited subset of the SDK, focused on signal generation and timing control through the Analog Out and Digital Out instruments. The WaveForms SDK must be installed separately.

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URL https://mass-spec.ru/projects/device_control/dwf4r/eng/

BugReports <https://github.com/andreysamokhin/dwf4r/issues>

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Contents

CarrierEnableDisable	2
CloseAllDevices	3
CloseDevice	4
ConfigureAnalogOut	4
GetAnalogOutIdle	5
GetAnalogOutInfo	6
GetAnalogOutMaster	7

GetAnalogOutRemaining	8
GetAnalogOutRepeatTrigger	9
GetAnalogOutStatus	9
GetAnalogOutTiming	10
GetAnalogOutTriggerSlope	11
GetAnalogOutTriggerSource	11
GetCarrierFunction	12
GetCarrierSettings	12
GetCarrierValues	13
GetDeviceTriggerInfo	14
IsCarrierEnabled	15
ListDevices	15
OpenDevice	16
ResetAnalogOut	17
SetAnalogOutIdle	18
SetAnalogOutMaster	18
SetAnalogOutRepeatTrigger	19
SetAnalogOutTiming	20
SetAnalogOutTriggerSlope	21
SetAnalogOutTriggerSource	22
SetCarrier	22
SetCarrierData	24
TriggerDevice	25

Index 26

CarrierEnableDisable *Enable or disable carrier output*

Description

Enable or disable the carrier node for an Analog Out channel.

EnableCarrier() enables the carrier node.

DisableCarrier() disables the carrier node.

Usage

```
EnableCarrier(device, channel)
```

```
DisableCarrier(device, channel)
```

Arguments

device A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.

channel An integer scalar representing the zero-based Analog Out channel index.

Details

These functions change the carrier node configuration. To apply the changes, call ApplyAnalogOutSettings() or StartAnalogOut().

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

CloseAllDevices	<i>Close all WaveForms devices opened by the current process</i>
-----------------	--

Description

Close all WaveForms device connections opened by the current R process. This operation is process-wide and is not limited to connections opened through **dwf4r**. It may therefore also close connections created by other packages or by direct use of the WaveForms SDK in the same R process. Connections owned by other processes are not affected.

After this function returns, all existing "dwf4r_device" objects and their copies are stale and cannot be used in subsequent **dwf4r** calls. Any WaveForms handles created by other code in the current R process are also invalidated.

Usage

```
CloseAllDevices()
```

Value

Invisibly returns NULL.

Examples

```
## Not run:
devices <- ListDevices()
if (nrow(devices) >= 1L) {
  device_sn <- devices$serial_number[[1L]]
  device <- OpenDevice(device_sn)
  CloseAllDevices()
}

## End(Not run)
```

CloseDevice	<i>Close a device</i>
-------------	-----------------------

Description

Close an active WaveForms device connection and mark the corresponding session as closed. After this function returns, the device object and any copies of it are stale and cannot be used in subsequent **dwf4r** calls.

Usage

```
CloseDevice(device)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
--------	---

Value

Invisibly returns NULL.

Examples

```
## Not run:
devices <- ListDevices()
if (nrow(devices) >= 1L) {
  device_sn <- devices$serial_number[[1L]]
  device <- OpenDevice(device_sn)
  CloseDevice(device)
}

## End(Not run)
```

ConfigureAnalogOut	<i>Configure Analog Out generation</i>
--------------------	--

Description

Apply settings, start, or stop Analog Out generation for a selected channel. Applying settings sends updated configuration to the device without changing the current channel state.

ApplyAnalogOutSettings() applies updated Analog Out settings.

StartAnalogOut() starts Analog Out generation.

StopAnalogOut() stops Analog Out generation.

Usage

```
ApplyAnalogOutSettings(device, channel)
```

```
StartAnalogOut(device, channel)
```

```
StopAnalogOut(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetAnalogOutIdle

Get the Analog Out idle output mode

Description

Retrieve the idle output mode currently configured for a selected Analog Out channel.

The idle mode controls the output while the generator is not actively running, including the Ready, Stopped, Done, and Wait states.

Usage

```
GetAnalogOutIdle(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A string containing "disable", "offset", "initial", or "hold".

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

`GetAnalogOutInfo`*Get Analog Out characteristics*

Description

Retrieve Analog Out characteristics of a device.

`GetAnalogOutChannelCount()` returns the number of Analog Out channels.

`GetAnalogOutChannelNodes()` returns the nodes supported by an Analog Out channel.

`GetAnalogOutRunRange()` returns the supported run-time range, in seconds, for an Analog Out channel.

`GetAnalogOutWaitRange()` returns the supported wait-time range, in seconds, for an Analog Out channel.

`GetAnalogOutRepeatRange()` returns the supported repeat-count range for an Analog Out channel.

`GetAnalogOutIdleModes()` returns the idle output modes supported by an Analog Out channel.

`GetAnalogOutNodeFunctionTypes()` returns the waveform functions supported by an Analog Out node.

`GetAnalogOutNodeFrequencyRange()` returns the supported frequency range for an Analog Out node.

`GetAnalogOutNodeAmplitudeRange()` returns the supported amplitude range for an Analog Out node.

`GetAnalogOutNodeOffsetRange()` returns the supported offset range for an Analog Out node.

`GetAnalogOutNodeSymmetryRange()` returns the supported symmetry range for an Analog Out node.

`GetAnalogOutNodePhaseRange()` returns the supported phase range, in degrees, for an Analog Out node.

`GetAnalogOutNodeSampleCountRange()` returns the supported range of sample counts for custom waveform data.

Usage

`GetAnalogOutChannelCount(device)`

`GetAnalogOutChannelNodes(device, channel)`

`GetAnalogOutRunRange(device, channel)`

`GetAnalogOutWaitRange(device, channel)`

`GetAnalogOutRepeatRange(device, channel)`

`GetAnalogOutIdleModes(device, channel)`

`GetAnalogOutNodeFunctionTypes(device, channel, node)`

`GetAnalogOutNodeFrequencyRange(device, channel, node)`

```
GetAnalogOutNodeAmplitudeRange(device, channel, node)
```

```
GetAnalogOutNodeOffsetRange(device, channel, node)
```

```
GetAnalogOutNodeSymmetryRange(device, channel, node)
```

```
GetAnalogOutNodePhaseRange(device, channel, node)
```

```
GetAnalogOutNodeSampleCountRange(device, channel, node)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
node	A string specifying the Analog Out node. One of "carrier", "fm", or "am".

Value

An integer, numeric, or character vector describing the requested Analog Out capability. Range functions return two elements containing the minimum and maximum supported values.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetAnalogOutMaster	<i>Get the Analog Out master channel</i>
--------------------	--

Description

Retrieve the state-machine master currently configured for a selected Analog Out channel.

Usage

```
GetAnalogOutMaster(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

An integer scalar containing the zero-based index of the configured master channel. If the returned index is equal to channel, the channel reports itself as its state-machine master. On Analog Discovery 2, this was observed experimentally to correspond to independent channel operation; this behavior is not explicitly described in the WaveForms SDK reference manual.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetAnalogOutRemaining *Get remaining Analog Out timing values*

Description

Retrieve the remaining run time or repeat count for a selected Analog Out channel. The remaining wait time cannot be retrieved because the WaveForms SDK does not provide a corresponding Analog Out wait-status function.

GetAnalogOutRemainingRun() returns the remaining run time of an Analog Out channel.

GetAnalogOutRemainingRepeat() returns the remaining repeat count of an Analog Out channel.

Usage

```
GetAnalogOutRemainingRun(device, channel, update = TRUE)
```

```
GetAnalogOutRemainingRepeat(device, channel, update = TRUE)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
update	A logical scalar indicating whether the Analog Out status should be updated before retrieving the remaining value. The default is TRUE. If FALSE, the value from the most recent Analog Out status update (performed with GetAnalogOutStatus()) is returned.

Details

The WaveForms SDK functions used to retrieve the remaining run time and repeat count do not themselves read the device. They return information obtained by the most recent GetAnalogOutStatus() (i.e. FDwfAnalogOutStatus()) call.

With the default update = TRUE, these functions update the Analog Out status before retrieving the remaining value. With update = FALSE, no status update is performed and the returned value may therefore be stale.

Setting update = FALSE can also be used to retrieve several values from the same status update. For example, first call GetAnalogOutStatus(), then call both remaining-value functions with update = FALSE.

Value

GetAnalogOutRemainingRun() returns a numeric scalar containing the remaining run time in seconds. GetAnalogOutRemainingRepeat() returns an integer scalar containing the remaining repeat count.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

`GetAnalogOutRepeatTrigger`*Get Analog Out repeat triggering*

Description

Retrieve whether a new trigger is included in each repeat cycle of a selected Analog Out channel.

Usage

```
GetAnalogOutRepeatTrigger(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A logical scalar. TRUE indicates that a new trigger is included in each wait-run repeat cycle; FALSE indicates that repeated cycles proceed without waiting for another trigger.

`GetAnalogOutStatus`*Get the state of an Analog Out channel*

Description

Retrieve the current state of a selected Analog Out channel.

Usage

```
GetAnalogOutStatus(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A string containing "ready", "armed", "wait", "running", "done", "config", "prefill", or "not_done". The value "unknown" is returned for an unrecognized SDK state.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetAnalogOutTiming	<i>Get Analog Out timing</i>
--------------------	------------------------------

Description

Retrieve the configured run time, post-trigger wait time, or repeat count for a selected Analog Out channel.

GetAnalogOutRun() returns the configured run length of an Analog Out channel.

GetAnalogOutWait() returns the configured delay between receiving a trigger and beginning signal generation.

GetAnalogOutRepeat() returns the configured repeat count of an Analog Out channel.

GetAnalogOutTiming() returns all configured timing parameters of an Analog Out channel.

Usage

```
GetAnalogOutRun(device, channel)
```

```
GetAnalogOutWait(device, channel)
```

```
GetAnalogOutRepeat(device, channel)
```

```
GetAnalogOutTiming(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
--------	---

channel	An integer scalar representing the zero-based Analog Out channel index.
---------	---

Details

These functions retrieve the configured timing values reported by the corresponding WaveForms SDK *Get functions. A run time of 0 represents continuous generation, a wait time of 0 represents no post-trigger delay, and a repeat count of 0 represents infinite repetition.

Value

GetAnalogOutRun() and GetAnalogOutWait() return a numeric scalar in seconds. GetAnalogOutRepeat() returns an integer scalar. GetAnalogOutTiming() returns a named list with the following elements:

run_time Numeric configured run time in seconds.

wait_time Numeric configured post-trigger wait time in seconds.

repeat_count Integer configured repeat count.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

`GetAnalogOutTriggerSlope`*Get the Analog Out trigger slope*

Description

Retrieve the trigger slope currently configured for a selected Analog Out channel.

Usage

```
GetAnalogOutTriggerSlope(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A string identifying the configured trigger slope.

`GetAnalogOutTriggerSource`*Get the Analog Out trigger source*

Description

Retrieve the trigger source currently configured for a selected Analog Out channel.

Usage

```
GetAnalogOutTriggerSource(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A string identifying the configured trigger source.

GetCarrierFunction	<i>Get the carrier waveform function</i>
--------------------	--

Description

Retrieve the waveform function currently configured for the carrier node of an Analog Out channel.

Usage

```
GetCarrierFunction(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A string identifying the configured waveform function. Possible values can be queried with GetAnalogOutNodeFunction

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetCarrierSettings	<i>Get carrier waveform settings</i>
--------------------	--------------------------------------

Description

Retrieve the current configuration of the carrier node for an Analog Out channel.

Usage

```
GetCarrierSettings(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

A named list with the following elements:

`enabled` Logical value indicating whether the carrier node is enabled.

`func` String identifying the configured waveform function.

`frequency` Numeric carrier frequency in Hz.

`amplitude` Numeric carrier peak amplitude in volts relative to the configured offset.

`offset` Numeric carrier offset in volts.

`symmetry` Numeric carrier symmetry in percent.

`phase` Numeric carrier phase in degrees.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetCarrierValues	<i>Get numeric carrier waveform parameters</i>
------------------	--

Description

Retrieve numeric parameters currently configured for the carrier node of an Analog Out channel.

`GetCarrierFrequency()` returns the configured carrier frequency in Hz.

`GetCarrierAmplitude()` returns the configured carrier peak amplitude in volts relative to the configured offset.

`GetCarrierOffset()` returns the configured carrier offset in volts.

`GetCarrierSymmetry()` returns the configured carrier symmetry in percent.

`GetCarrierPhase()` returns the configured carrier phase in degrees.

Usage

```
GetCarrierFrequency(device, channel)
```

```
GetCarrierAmplitude(device, channel)
```

```
GetCarrierOffset(device, channel)
```

```
GetCarrierSymmetry(device, channel)
```

```
GetCarrierPhase(device, channel)
```

Arguments

`device` A "dwf4r_device" object returned by `OpenDevice()` that refers to a device connection that is still open.

`channel` An integer scalar representing the zero-based Analog Out channel index.

Details

The functions return values reported by the corresponding WaveForms SDK *Get functions. A returned value may differ slightly from the value supplied to a setter if the SDK adjusted it to a value supported by the device.

Value

A numeric scalar containing the requested carrier parameter.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

GetDeviceTriggerInfo *Get device trigger capabilities*

Description

Retrieve trigger capabilities of a WaveForms device.

GetDeviceTriggerSources() returns the trigger sources supported by the device and available through the current **dwf4r** interface.

Possible values include "none", "pc", "detector_analog_in", "detector_digital_in", "analog_in", "digital_in", "digital_out", "analog_out_1", "analog_out_2", "external_1", and "external_2".

GetDeviceTriggerSlopes() returns the trigger slopes supported by the device. Possible values are "rising", "falling", and "either".

Usage

```
GetDeviceTriggerSources(device)
```

```
GetDeviceTriggerSlopes(device)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
--------	---

Value

A character vector containing supported trigger source or slope names, depending on the function.

IsCarrierEnabled	<i>Check whether carrier output is enabled</i>
------------------	--

Description

Retrieve the configured enable state of the carrier node for an Analog Out channel.

Usage

IsCarrierEnabled(device, channel)

Arguments

- | | |
|---------|---|
| device | A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open. |
| channel | An integer scalar representing the zero-based Analog Out channel index. |

Value

A logical scalar, TRUE if the carrier node is enabled and FALSE if it is disabled.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

ListDevices	<i>List detected devices</i>
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Description

Enumerate detected devices using the filters specified by filters and retrieve some additional information for each device.

Usage

ListDevices(filters = "all")

Arguments

- | | |
|---------|--|
| filters | A character vector specifying one or more enumeration filters. The default, "all", corresponds to the SDK's default enumeration filter. Other supported values are "usb", "network", "axi", "remote", "audio", and "demo". |
|---------|--|

Value

A data frame with one row per enumerated device and the following columns:

`device_index` Zero-based index of the device in the current SDK enumeration result.
`device_id` Integer device-type identifier defined by the SDK.
`device_type` Character label mapped from `device_id`; unknown SDK identifiers are reported as "Unknown".
`device_revision` Integer device-revision identifier.
`device_name` Character string containing the device name.
`user_name` Character string containing the user-assigned device name.
`serial_number` Character string containing the unique device serial number.
`is_opened` Logical value indicating whether the device is already opened by the current or another process.
`config_count` Integer number of available configurations for the device.

Examples

```
## Not run:
devices <- ListDevices()
print(devices)

usb_devices <- ListDevices(filters = "usb")
print(usb_devices)

## End(Not run)
```

OpenDevice

Open a device

Description

Open a WaveForms device, register the connection in the current R session, and return an object that can be passed to other **dwf4r** functions.

Usage

```
OpenDevice(serial_number = NULL)
```

Arguments

`serial_number` A string containing the serial number of the device to open. If NULL, the device is selected automatically when exactly one compatible device is detected.

Details

The device is opened using its default configuration. Automatic configuration in the WaveForms SDK is disabled to reduce the communication overhead of individual setting calls. Instrument settings are therefore applied explicitly by the corresponding **dwf4r** configuration functions. This behavior is managed internally and is not currently user-configurable.

Value

An opaque object of class "dwf4r_device" representing the opened connection. Pass this object unchanged to other **dwf4r** functions.

Examples

```
## Not run:
devices <- ListDevices()

if (nrow(devices) >= 1L) {
  if (nrow(devices) == 1L) {
    device <- OpenDevice()
  } else {
    device_sn <- devices$serial_number[[1L]]
    device <- OpenDevice(device_sn)
  }

  print(device)

  CloseDevice(device)
}

## End(Not run)
```

ResetAnalogOut	<i>Reset Analog Out channel</i>
----------------	---------------------------------

Description

Reset the Analog Out settings of a selected channel to their SDK defaults. Because automatic configuration is disabled by **dwf4r**, call `ApplyAnalogOutSettings()` or `StartAnalogOut()` to apply the reset configuration to the device.

Usage

```
ResetAnalogOut(device, channel)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

SetAnalogOutIdle	<i>Set Analog Out idle output</i>
------------------	-----------------------------------

Description

Set the idle output mode for a selected Analog Out channel.

The idle output mode controls the Analog Out output while the generator is not actively running, including Ready, Stopped, Done, and Wait states.

Usage

```
SetAnalogOutIdle(device, channel, idle)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
idle	A string specifying the idle output mode. Supported values are "disable", "offset", "initial", and "hold".

Details

The "offset" mode uses the configured offset level. Use SetCarrierOffset() to set that voltage level. If device auto-configuration is disabled, call ApplyAnalogOutSettings() or StartAnalogOut() after changing Analog Out settings.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

SetAnalogOutMaster	<i>Set the Analog Out master channel</i>
--------------------	--

Description

Set the state-machine master for a selected Analog Out channel.

Usage

```
SetAnalogOutMaster(device, channel, master_channel)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
master_channel	An integer scalar representing the zero-based Analog Out channel index to use as the state-machine master.

Details

A channel whose master differs from itself is controlled by the state machine of the master channel. Trigger, wait, run, and repeat settings therefore normally need to be configured only for the master channel.

Setting master_channel equal to channel appears consistent with independent channel operation. Although this behavior is not explicitly described in the WaveForms SDK reference manual, it was observed experimentally with Analog Discovery 2.

This function updates the pending Analog Out configuration. Because automatic configuration is disabled by **dwf4r**, call ApplyAnalogOutSettings() to apply the setting without changing the current channel state, or StartAnalogOut() to apply the setting.

When starting synchronized channels, initialize the slave channel before starting the master channel.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

SetAnalogOutRepeatTrigger

Set Analog Out repeat triggering

Description

Set whether a new trigger is included in each repeat cycle of a selected Analog Out channel.

Usage

```
SetAnalogOutRepeatTrigger(device, channel, repeat_trigger)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
repeat_trigger	A logical scalar. If TRUE, a new trigger is included in each wait-run repeat cycle. If FALSE, repeated cycles proceed without waiting for another trigger.

Details

This setting controls trigger behavior during repetition and does not specify the number of repetitions. Use SetAnalogOutRepeat() to set the repeat count.

This function updates the pending Analog Out configuration. Because automatic configuration is disabled by **dwf4r**, call ApplyAnalogOutSettings() to apply the setting without changing the current channel state, or StartAnalogOut() to apply the setting and start generation.

Value

Invisibly returns NULL.

SetAnalogOutTiming	<i>Set Analog Out timing</i>
--------------------	------------------------------

Description

Set the run time, post-trigger wait time, or repeat count for a selected Analog Out channel.

SetAnalogOutRun() sets the run length of an Analog Out channel.

SetAnalogOutWait() sets the delay between receiving a trigger and beginning signal generation.

SetAnalogOutRepeat() sets the repeat count of an Analog Out channel.

SetAnalogOutTiming() sets one or more timing parameters of an Analog Out channel.

Usage

```
SetAnalogOutRun(device, channel, run_time)
```

```
SetAnalogOutWait(device, channel, wait_time)
```

```
SetAnalogOutRepeat(device, channel, repeat_count)
```

```
SetAnalogOutTiming(
    device,
    channel,
    run_time = NULL,
    wait_time = NULL,
    repeat_count = NULL
)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
run_time	A numeric scalar specifying the run length in seconds. A value of 0 requests continuous generation.
wait_time	A numeric scalar specifying the delay, in seconds, between receiving a trigger and beginning signal generation. A value of 0 disables this delay.
repeat_count	An integer scalar specifying the repeat count. A value of 0 requests infinite repetition.

Details

These functions update the pending Analog Out configuration. Because automatic configuration is disabled by **dwf4r**, call `ApplyAnalogOutSettings()` to apply the settings without changing the current channel state, or `StartAnalogOut()` to apply the settings and start generation.

In `SetAnalogOutTiming()`, a NULL argument leaves the corresponding setting unchanged. Selected parameters are updated in the order `run_time`, `wait_time`, and `repeat_count`. If an error occurs, parameters set before the error remain in the pending configuration.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

SetAnalogOutTriggerSlope

Set the Analog Out trigger slope

Description

Set the trigger slope for a selected Analog Out channel.

Usage

```
SetAnalogOutTriggerSlope(device, channel, slope)
```

Arguments

device	A "dwf4r_device" object returned by <code>OpenDevice()</code> that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
slope	A string specifying the trigger slope. Recognized values are "rising", "falling", and "either".

Details

This function updates the pending Analog Out configuration. Because automatic configuration is disabled by **dwf4r**, call `ApplyAnalogOutSettings()` to apply the setting without changing the current channel state, or `StartAnalogOut()` to apply the setting and start generation.

Value

Invisibly returns NULL.

SetAnalogOutTriggerSource

Set the Analog Out trigger source

Description

Set the trigger source for a selected Analog Out channel.

Usage

```
SetAnalogOutTriggerSource(device, channel, source)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
source	A string specifying the trigger source. Recognized values include "none", "pc", "detector_analog_in", "detector_digital_in", "analog_in", "digital_in", "digital_out", "analog_out_1", "analog_out_2", "external_1", and "external_2". This list is not exhaustive; available sources depend on the selected device and can be queried with GetDeviceTriggerSources().

Details

This function updates the pending Analog Out configuration. Because automatic configuration is disabled by **dwf4r**, call ApplyAnalogOutSettings() to apply the setting without changing the current channel state, or StartAnalogOut() to apply the setting and start generation.

Value

Invisibly returns NULL.

SetCarrier

Set carrier waveform parameters

Description

Set parameters of the carrier node for an Analog Out channel.

SetCarrierFunction() sets the waveform function of the carrier node.

SetCarrierFrequency() sets the frequency of the carrier node.

SetCarrierAmplitude() sets the amplitude of the carrier node.

SetCarrierOffset() sets the offset of the carrier node.

SetCarrierSymmetry() sets the symmetry of the carrier node.

SetCarrierPhase() sets the phase (in degrees) of the carrier node.

SetCarrier() sets one or more parameters of the carrier node for an Analog Out channel. Carrier parameters are updated sequentially. If an error occurs, parameters set before the error remain in the pending configuration but are not applied until ApplyAnalogOutSettings() or StartAnalogOut() is called.

Usage

```

SetCarrierFunction(device, channel, func)

SetCarrierFrequency(device, channel, frequency)

SetCarrierAmplitude(device, channel, amplitude)

SetCarrierOffset(device, channel, offset)

SetCarrierSymmetry(device, channel, symmetry)

SetCarrierPhase(device, channel, phase)

SetCarrier(
    device,
    channel,
    func = NULL,
    frequency = NULL,
    amplitude = NULL,
    offset = NULL,
    symmetry = NULL,
    phase = NULL
)

```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
func	A string specifying the carrier waveform function. Available functions depend on the selected device and can be queried with GetAnalogOutNodeFunctionTypes(). The list of functions from the manual: • "dc" - DC value set as offset; • "sine" - sine waveform; • "square" - square waveform (offset +/- amplitude); • "triangle" - triangle waveform; • "ramp_up" - waveform with a ramp-up voltage at the beginning; • "ramp_down" - a waveform with a ramp-down voltage at the end; • "noise" - noise waveform from random samples; • "pulse" - pulse waveform (offset + amplitude); • "trapezium" - trapezium; • "sine_power" - sine with symmetry used as power function; • "sine_na" - sine waveform for Network Analyzer, dynamic frequency adjustment without glitch • "dual_custom" - double buffering of the waveform in the device for smooth switching. • "dual_pattern" - double buffering of the waveform in the device for smooth switching. • "custom_pattern" - waveform from custom repeated data with fixed sample rate;

	• "play_pattern" - waveform in stream play style (it provides constant sample rate); • "custom" - waveform from custom samples (optimizes for average requested frequency, sample output lengths may vary by one system frequency period); • "play" - waveform in stream play style (optimizes for average requested frequency).
frequency	A numeric scalar specifying the frequency in Hz.
amplitude	A numeric scalar specifying the carrier amplitude, in volts. For symmetric waveforms such as a sine wave, this is the voltage excursion from the configured offset to either peak. The output therefore ranges from offset - amplitude to offset + amplitude, corresponding to a peak-to-peak voltage of 2 * amplitude.
offset	A numeric scalar specifying the carrier offset, in volts. For symmetric waveforms, the offset is the midpoint between the minimum and maximum output voltages.
symmetry	A numeric scalar specifying the symmetry in percent.
phase	A numeric scalar specifying the carrier phase in degrees.

Details

These functions set carrier parameters only. Use ApplyAnalogOutSettings() to send updated settings to the device without changing the current channel state, or StartAnalogOut() to apply settings and start generation.

The requested value may differ from the value actually configured by the device; use the corresponding GetCarrier*() function to retrieve the actual value.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

SetCarrierData	Set carrier custom waveform samples
----------------	-------------------------------------

Description

Upload a custom sample vector for the carrier node of an Analog Out channel. Samples must be normalized to the range [-1, 1].

This function uploads sample data but does not select the custom waveform function or start generation. Use SetCarrierFunction(..., func = "custom") and then call ApplyAnalogOutSettings() or StartAnalogOut().

Usage

SetCarrierData(device, channel, data)

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
channel	An integer scalar representing the zero-based Analog Out channel index.
data	A numeric vector of waveform samples normalized to the range $[-1, 1]$.

Value

Invisibly returns NULL.

See Also

See the vignette *Basic Analog Out Functionality* for a complete Analog Out workflow.

TriggerDevice	<i>Trigger a WaveForms device</i>
---------------	-----------------------------------

Description

Generate a PC trigger pulse for a WaveForms device.

Instruments configured to use the "pc" trigger source can respond to this pulse.

Usage

```
TriggerDevice(device)
```

Arguments

device	A "dwf4r_device" object returned by OpenDevice() that refers to a device connection that is still open.
--------	---

Value

Invisibly returns NULL.

Index

ApplyAnalogOutSettings
(ConfigureAnalogOut), 4

CarrierEnableDisable, 2

CloseAllDevices, 3

CloseDevice, 4

ConfigureAnalogOut, 4

DisableCarrier (CarrierEnableDisable), 2

EnableCarrier (CarrierEnableDisable), 2

GetAnalogOutChannelCount
(GetAnalogOutInfo), 6

GetAnalogOutChannelNodes
(GetAnalogOutInfo), 6

GetAnalogOutIdle, 5

GetAnalogOutIdleModes
(GetAnalogOutInfo), 6

GetAnalogOutInfo, 6

GetAnalogOutMaster, 7

GetAnalogOutNodeAmplitudeRange
(GetAnalogOutInfo), 6

GetAnalogOutNodeFrequencyRange
(GetAnalogOutInfo), 6

GetAnalogOutNodeFunctionTypes
(GetAnalogOutInfo), 6

GetAnalogOutNodeOffsetRange
(GetAnalogOutInfo), 6

GetAnalogOutNodePhaseRange
(GetAnalogOutInfo), 6

GetAnalogOutNodeSampleCountRange
(GetAnalogOutInfo), 6

GetAnalogOutNodeSymmetryRange
(GetAnalogOutInfo), 6

GetAnalogOutRemaining, 8

GetAnalogOutRemainingRepeat
(GetAnalogOutRemaining), 8

GetAnalogOutRemainingRun
(GetAnalogOutRemaining), 8

GetAnalogOutRepeat
(GetAnalogOutTiming), 10

GetAnalogOutRepeatRange
(GetAnalogOutInfo), 6

GetAnalogOutRepeatTrigger, 9

GetAnalogOutRun (GetAnalogOutTiming), 10

GetAnalogOutRunRange
(GetAnalogOutInfo), 6

GetAnalogOutStatus, 9

GetAnalogOutTiming, 10

GetAnalogOutTriggerSlope, 11

GetAnalogOutTriggerSource, 11

GetAnalogOutWait (GetAnalogOutTiming),
10

GetAnalogOutWaitRange
(GetAnalogOutInfo), 6

GetCarrierAmplitude (GetCarrierValues),
13

GetCarrierFrequency (GetCarrierValues),
13

GetCarrierFunction, 12

GetCarrierOffset (GetCarrierValues), 13

GetCarrierPhase (GetCarrierValues), 13

GetCarrierSettings, 12

GetCarrierSymmetry (GetCarrierValues),
13

GetCarrierValues, 13

GetDeviceTriggerInfo, 14

GetDeviceTriggerSlopes
(GetDeviceTriggerInfo), 14

GetDeviceTriggerSources
(GetDeviceTriggerInfo), 14

IsCarrierEnabled, 15

ListDevices, 15

OpenDevice, 16

ResetAnalogOut, 17

SetAnalogOutIdle, 18

SetAnalogOutMaster, 18

SetAnalogOutRepeat
(SetAnalogOutTiming), 20

SetAnalogOutRepeatTrigger, 19

SetAnalogOutRun (SetAnalogOutTiming), 20

SetAnalogOutTiming, 20

SetAnalogOutTriggerSlope, 21

SetAnalogOutTriggerSource, [22](#)
SetAnalogOutWait (SetAnalogOutTiming),
 [20](#)
SetCarrier, [22](#)
SetCarrierAmplitude (SetCarrier), [22](#)
SetCarrierData, [24](#)
SetCarrierFrequency (SetCarrier), [22](#)
SetCarrierFunction (SetCarrier), [22](#)
SetCarrierOffset (SetCarrier), [22](#)
SetCarrierPhase (SetCarrier), [22](#)
SetCarrierSymmetry (SetCarrier), [22](#)
StartAnalogOut (ConfigureAnalogOut), [4](#)
StopAnalogOut (ConfigureAnalogOut), [4](#)

TriggerDevice, [25](#)